

AN ALTERNATE HISTORY OF THE FIRST COMPUTER AGE

The Book of Conditions

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Three people. One finite apparatus.
The conditions that make a calculation trustworthy become the conditions under which people
must work.
Literary alternate history; not a historical reconstruction.

I. THE RETURNED TABLE

The packet came back tied with the office's own green tape.

Edward Halden knew the tape because he had ordered twelve dozen yards of it and argued down the price. He knew the paper because its weight had been entered in the delivery account. He knew the seal of Guardian Assurance, pressed too hard into the red wax, and Griffith Davies's narrow hand upon the wrapper. What he did not know was why the packet should have returned unopened at one end and cut cleanly along the other.

The calculating room continued behind him. A shaft turned beneath the floor. Every few seconds the card reader lifted, admitted a stiff rectangle, and let it fall into the receiving box. The apparatus did not sound like thought. It sounded like six metal cupboards being locked in sequence.

Davies's letter contained three sentences.

The figures in the enclosed table did not agree with Guardian's own calculation, by hand, of the ages from forty-seven onward, which Davies kept as a check on every table he bought. The office was to suspend the related work. Mr Davies requested the source sheets, the operation cards, both checking signatures, and an account of the discrepancy under clause eleven.

That explained the cut. Davies had opened the packet along its far edge so that the office's own tape and seal would come back as they had gone out: evidence, not courtesy.

Halden read the letter twice and placed it flat upon his desk. He did not call for Babbage. He did not yet call for the clerk who had prepared the cards. First he opened the returned table.

The error was not theatrical. No column descended into nonsense. The values remained orderly and plausible; each differed from Guardian's by an amount small enough to survive an impatient glance and large enough to alter a later reserve. That was worse. An absurd answer accused itself. A reasonable wrong answer recruited the reader.

He took the configuration register from the locked cabinet and checked the seal. Unbroken. The mechanism described there was still the mechanism in the room: six signed registers, addition and subtraction, the same transfer order, the same limits they had stated three years before. He found the run trace. It showed no jam, correction, or repeated card. The second run agreed with the first.

By noon Babbage stood at the long table with his spectacles in one hand and the returned sheets in the other.

"The recurrence is exact," he said.

"The table is wrong."

"Wrong, yes; but observe where." Babbage set down his spectacles and ran one finger along the trace. "The first value supplied for forty-seven, and nowhere after it. Every transfer that follows is true. No wheel introduced this. The Engine carried, as it was built to carry, precisely what it was given."

Halden knew that relief. He had felt it himself on finding the seal unbroken. Now Babbage was looking only at the trace.

"We gave it," Halden said.

"The preparation desk gave it."

"That desk is in this office."

Babbage picked up his spectacles. "Then correct the copy and repeat."

Halden looked through the glass partition. The preparers sat at facing desks, reducing customer sheets to a form the cards could carry. Beyond them, the apparatus continued its patient sequence. To a visitor the machine was the marvel and the clerks were furniture. The accounts said otherwise. For every hour of its operation, people spent several hours deciding what it should be given, and more deciding whether its answer could leave the room.

He sent for the Guardian source sheet and the preparation copy. At a quarter past one, he found the first divergence. Guardian had supplied 3,742. The preparation copy read 3,472. The two middle figures had exchanged places in a hand so familiar that Halden recognized it before he read the initials.

Samuel Wills came in pale. He had brought the ruler and pencil from his desk although Halden had asked only for him; he kept them aligned between his hands while he stood.

"Did you compare this line with the source?" Halden asked.

"I read the copy back, sir."

“To whom?”

The young man looked towards the checking room.

“Mr Neale had the Lothbury packet. I read it back to myself.”

“That is not a comparison.”

“No, sir.”

“Who instructed you to release the cards?”

The clerk said nothing. The ruler shifted in his hands, and he brought its end level with the pencil again. Halden had been about to ask the question again, louder. He dismissed him instead.

The young man took both tools away. Through the partition Halden watched him lay them square with the edge of his desk before sitting down. Then Halden opened the delivery ledger.

There should have been two marks beside the Guardian job: the preparer’s and an independent checker’s. There was one. Beneath it, in the margin reserved for an exception, was a short authorization in Halden’s own hand.

Second comparison to follow. Delivery permitted to preserve date.

E. H.

He had not forgotten the morning. The wages had been due on Friday. Guardian’s delivery date fell that day; missing it would make its one hundred and eighty pounds returnable. Neale had been checking the Lothbury office’s table, whose delay would also have cost the office money. Halden had told himself that a single delivery could borrow two days from its own conditions and repay them on Monday.

He had seen the other choices. He could have asked Davies for an extension and admitted that the office had sold a date it could not safely keep. He could have moved Neale from the Lothbury packet and broken that promise instead. Or he could let one signature stand temporarily for two and preserve both appearances until Monday. The third course had felt least like a decision because it moved no visible object and required no difficult letter. Only a line in the margin had changed.

Monday had filled itself before it arrived.

Babbage read the authorization without sitting down.

“It remains a human failure,” he said. “The apparatus received three thousand four hundred and seventy-two and treated it with perfect fidelity. To charge the mechanism with the table is to confuse obedience to an operation with judgment upon its premise. I have met that confusion before, Mr Halden, in a member of the Upper House. I had not expected to meet it here.”

“Of course it is a human failure.”

The answer surprised him. Babbage turned.

“Wills transposed two figures,” Halden continued. “I waived Neale’s comparison. We sold four dates with hands for two. Which entry do you want first?”

“You have passed from one supplied value to the constitution of the office. That may be a second inquiry; it is not a failure of the calculating mechanism.”

“We took one witnessed trial and sold four tables upon it.”

For a moment neither man spoke. Through the partition came the fall of another card.

On the shelf above Halden’s desk stood the test packet from 1851, its seals cut but preserved. Beside it lay Lady Lovelace’s qualification on capacity. He had filed them together because their paper was the same size. Until that morning he had not admitted that they were parts of the same result.

Davies had asked for both checking signatures. Halden dipped his pen and began the reply with the one fact no calculation could improve.

There was no second signature.

II. THIRTY DAYS

Eleven years earlier, in August 1843, Charles Babbage received Ada Lovelace’s proposal as one receives a map drawn by another person of one’s own house: with admiration for the accuracy and indignation at the implied right to enter.

The paper had arrived after the Notes were finished, after the Turin account had been turned inside out and enlarged until Menabrea’s sketch seemed almost an introduction to her footnotes. Babbage had praised

the work freely. Praise cost him nothing that diminished the Engine. The proposal was different. It asked for powers, money, duties, and a division of authority around an object that existed most completely in his mind.

He read it on the fifteenth of August and again the next morning. On the eighteenth, Lovelace came to his house in Dorset Street.

Models and loose drawings occupied the surfaces that books had not conquered. A segment of mechanism stood near the window, its polished faces catching a weak summer light. Lovelace did not go to it at once. She waited while Babbage squarely aligned the pages of her proposal, though they had arrived numbered.

She had spent too much of her life being received as somebody's relation: Lord Byron's daughter, Lady Byron's problem, William's wife, Babbage's gifted young interpreter. The Notes had altered that relation without abolishing it. People who admired what she had written still found ways to make the writing an emanation of the machine and the machine an emanation of the man before her. The proposal was her attempt to make admiration incur an obligation. She knew how readily that could be described as ambition in a woman and governance in a man. Knowing did not make the risk smaller.

"You have written a constitution around the Treasury's refusal," he said, "and called it an answer to the Engine. The Treasury answered me in fewer words. I have not forgiven it the economy."

"Then it should be easy to refuse," she said.

"You suppose, if I follow you, that because the government would not continue, a committee of private persons may be made wiser by clauses. I have sat on committees, Lady Lovelace. I have not seen the clauses improve them."

"No. I suppose that if one person holds the money, a second the design, and a third the right to refuse a claim, then each becomes a function of the other two, and none of us can be solved alone."

He moved towards the model. "The machine cannot be governed as if its proper form were already known. I did not know it in 1834, I do not know it this morning, and I shall not know it on the day your committee requires me to swear to it."

"If it is to be paid for," she said, "it must be governed as if somebody can say what is being paid for."

He had made much the same argument against Clement and against the Treasury. He lifted a wheel from the table. His thumb found a nick in one tooth and stopped there.

"And whatever I sign becomes permanent the moment the ink is dry. I learned that from Clement, who kept the tools and called it the custom of his trade. If I sign an inferior arrangement, you will have protected the undertaking by preventing the Engine."

"Then do not sign an arrangement whose inferiority you can already see. Sign the method by which an arrangement is proposed, tried, and refused without carrying the undertaking down with it."

"With you holding the purse?"

"With no purse at all, at present. I have been perfectly explicit about that."

That checked him. The document had the architecture of command without its means. Lovelace could name conditions, and he could name mechanisms, but the proposal contained no person whose ordinary occupation was to convert either into prices, stages, suppliers, and liabilities. He had taken that absence as evidence that the whole scheme was unreal. Now he saw that it was simply an office left empty.

"You have omitted the man who can say no," he said.

She turned back through the numbered pages. The office was not there.

"I had assigned that labour to you."

"Without asking me."

"Yes."

He laughed despite himself.

Lovelace approached the model at last. "Do not accept my proposal," she said. "Bind us only to the question: is there one finite obligation which does not falsify the general design?"

"And if there is not?"

"Then write the refusal and its reason. Any later proposal must answer it before anyone is asked to cast a metaphor in brass."

Babbage squared the pages again. Between accepting authority he had not tested and repeating the refusal he already knew, he found a smaller unit.

"Thirty days," he said.

"For what?"

“To find your missing man. Or to establish that he ought to remain missing.”

She held out her hand. “And during those thirty days?”

“Nothing is built. Nothing is subscribed. Nothing is announced.”

“And the refusal, if it comes, is written with its reason.”

He hesitated before taking her hand, not because the condition was large, but because it was small enough to bind him.

“Written,” he said.

III. THE ESTIMATOR

Edward Halden first encountered the Analytical Engine as eleven pounds of paper delivered by a boy from George Farrow’s Lambeth works who would not wait for an answer.

The packet contained drawings at several scales, explanatory memoranda in two hands, lists of operations, and a note from Farrow. Halden knew him through instrument and railway supply; Farrow knew Babbage through men who had worked in the old Clement network. The note was apologetic. Mr Babbage required not a price, it said, but an opinion as to whether a price might be made. Farrow had forwarded the inquiry rather than answer it because his works might later bid for whatever object survived the answer. He would not be permitted to certify his own possibility.

Halden weighed the packet because the absurdity of eleven pounds amused him. Then he read until midnight.

He had spent twenty years learning that a drawing could lie without containing a false line. It lied by silence: about the tool that did not yet exist, the surface another shop must finish, the order in which two correct things became one unusable thing, the week in which wages continued while a decision did not. The drawings in the packet were more truthful than most and more dangerous. They showed a mind able to answer questions no contractor had yet been asked, surrounded by questions no drawing could answer for him.

The next afternoon, the twenty-sixth of Babbage’s thirty days, he met Babbage and Lovelace in Dorset Street. Babbage began with the general architecture. Halden let him speak for twenty minutes, then placed a blank sheet over the largest drawing.

“Which object am I to price?” he asked.

Babbage looked at the sheet as if Halden had draped the Engine for its funeral. “The practicability of the Engine.”

“I can’t put practicability on a bill, sir. The man who pays it will want to know what it weighs.”

“The Engine can be resolved into orders of mechanism: the store, the mill, the reading of the cards, the printing, and so down. One of those orders might be priced, I suppose, without anyone mistaking it for the whole, though I have seldom met the man who would not.”

“Which one? Tell me where it stops. I can’t price a thing without an edge.”

Lovelace turned away to hide a smile. Babbage saw it.

“You have rehearsed this.”

“We have not met,” Halden said.

“Then you possess the same defect independently.”

Halden uncovered the drawing. “I can give you four weeks. At the end you will have four lists: what can be priced; what a works must review; what no present price can carry; and what result permits a customer to say yes or no. The last line may be: build nothing.”

“And your fee for the possibility of refusal?”

“Twenty-five pounds.”

It was more than Babbage had expected to pay for an opinion and less than Halden could afford to waste. His eldest child would begin school that autumn. Two railway accounts remained unpaid. If he accepted a vague commission, every later hour would be disputed; if he refused it outright, another man might promise what could not be delivered and be paid first.

Lovelace asked, “What authority would the fee purchase?”

“None beyond the report. If you continue afterward, you must decide who may alter a promised configuration, who holds money, who judges manufacture, who accepts the result, and what occurs after a failed test. If the same person holds all five, the paper is theatre.”

Babbage walked to the window. “You speak as if division produced competence.”

“It doesn’t make anybody competent. It tells me whose hours go on the bill, and whose word stops the wages.”

Babbage paid the fee that evening from his own purse, four days inside his term. No refusal was written.

For six evenings Halden tried to bound the Engine by selecting a portion of its proposed use, and each boundary opened upon machinery whose final form remained elsewhere.

The reversal came after he had covered the largest drawing with unpaid accounts. Only the edges remained: a shaft at one corner, a run of figures at another, arrows entering the paper and not returning. That night he put the drawing away. He set one of Lovelace’s worked operations beside a blank bill of quantities and began with the answer instead.

For the first time the list grew shorter as he worked. A register could be counted. A card could be prepared. A recurrence could be calculated elsewhere and sealed. Conditional branches went back into Babbage’s packet; printing went with them. Near dawn Halden found himself smiling at six empty columns on a sheet of paper. He numbered them twice, blotted the first row, and carried the sheet towards the window before noticing that the street was still dark.

The Notes supplied the route. Cards might direct operations upon symbols; the full consequence was immense, but the smallest material question was not. Could a fixed set of registers take values, follow an externally prepared sequence of additions and subtractions, and reproduce an independently known recurrence?

Six registers. Six decimal digits. Thirty card steps. Manual entry and transcription. No conditional branch. No automatic printing. No claim that the object was the Analytical Engine.

When Halden presented the proposal, Babbage read the exclusions first.

“You have defined the apparatus by subtracting from it every power that makes the architecture worth defining. It is an engine as a keyhole is a house.”

“I’ve defined what another man can be asked to finish.”

“And the rest?”

“The rest is entered here.” Halden placed a second, thinner book beside the report. “Every excluded improvement and dependency. Make it as general as the truth requires. It cannot draw one shilling from the apparatus I have priced.”

Lovelace opened the thin book. Its pages were numbered before they had content.

“And if the little apparatus succeeds?” she asked.

“Then it succeeds at the question we gave it.”

“People will say more. I have had more said about me all my life.”

Halden looked at Babbage, who had already begun correcting one of the register arrangements in pencil. Lovelace drew a recurrence down the margin and passed the sheet back. Babbage altered the third operation, objected to the six-digit limit, and then forgot the objection while repairing the fifth.

They worked until the room had to be lit. Halden crossed out an automatic return, the cards backing to repeat a sequence, because he could not price the machinery it required. Babbage began another line below the thirtieth operation. Lovelace moved his pencil to the first numbered page of the successor register. He looked at her hand, then continued writing there.

IV. WHAT CAN BE BOUGHT

Griffith Davies did not object to the Engine. He objected to being asked to have an opinion of it.

“If Mr Babbage has made a new principle,” he told Halden, “then I wish him every honour due to principles. Guardian purchases calculations.”

They sat beneath shelves of mortality tables in a room where every abstraction had eventually acquired a price. Davies had agreed to see Halden because Babbage’s earlier work on assurance was known to him, and

because Lady Lovelace's Notes had made the proposed operations less opaque than the drawings. Neither reason amounted to a contract.

Halden laid out the six-register proposal. Davies kept one finger in his copy of Lovelace's Notes. Several corners had been folded down; beside one operation he had written a reserve consequence in pencil.

"What calculation?" Davies asked.

"One from your ordinary work. You keep the source and the answer. We receive sealed copies."

"You have not chosen it."

"No."

Davies closed the Notes upon his finger. "Then at present you are pricing a machine without a use."

He turned the pages of the proposal and paused at the exclusions, as Halden had known he would. "No printing."

"A clerk transcribes the result."

"No conditional operation."

"Not in the accepted configuration."

"And after thirty steps?"

"A person prepares another sequence."

Davies closed the proposal. "Then do not call it an Engine in anything you send to my directors."

It was the first condition on which they agreed.

Guardian possessed a useful inconvenience: a set of recurrence calculations whose inputs could be sealed and whose answers could be produced by Guardian's own computers, working by hand, without reference to Babbage's mechanism. Davies selected twelve, not because they were the most impressive but because disagreement could be located. Guardian would pay for a feasibility comparison and reserve a further payment for a witnessed test. It would acquire no share of the architecture and promise no recurring business.

"If your apparatus differs from our figures?" Davies asked.

"We locate the difference."

"Before or after I pay?"

Davies had made a profession from promises that matured after the people who sold them were gone. Halden glanced at the folded corners of the Notes. Davies had followed the operations far enough to write in the margin. He had reopened none of those pages to settle the payment.

"Before final payment," Halden said. "You keep the right to reject it. The protocol must still tell us whether the difference entered with your source, our preparation, the mechanism, our transcription, or your comparison."

Davies repeated "protocol" as though it were a foreign coin, and reopened the proposal.

Guardian had refused to buy an Engine. The second refusal arrived from Manchester in a wooden box.

Halden had sent drawings of the reduced interfaces to Whitworth's works with a request for a review, not a blessing. The returned box contained gauges, three annotated sheets, and an account. Several surfaces could be made and inspected within ordinary practice. Two tolerances were expressed more finely than the drawings could justify. One shared drive could not be assessed until the loads and assembly order were fixed. The proposed date was fiction.

Babbage read the report as an accusation of temporary ignorance. Halden read it as a list of conditions. Lovelace, at Ockham, received copies of both the Guardian memorandum and the Manchester review with a letter asking which objections were fatal.

Her answer returned four days later.

None, she wrote, if they were allowed to alter the promise. All, if the promise were allowed to remain vague.

She divided the work into two columns. On one side were things the apparatus must do: hold six values, obey a finite card sequence, perform addition and subtraction, repeat twelve declared cases, and expose a trace. On the other were things the undertaking must do: preserve inputs, distinguish preparation from operation, compare independently, attribute disagreement, state exclusions, and stop if a stage failed.

Babbage objected that the second column contained no machinery.

"It contains the reason anyone may believe the machinery," she answered when they next met. "Without it you have a very beautiful clock that nobody is obliged to consult."

Halden placed Davies's memorandum beside Whitworth's report. For the first time, he could draw a line beneath them and write a total.

V. THE FINITE COMPACT

They founded the undertaking by deciding who could stop it.

This was Halden's preferred account. Babbage called it unnecessarily negative. Lovelace asked to see the stopping clauses before the title page.

The agreement occupied sixteen pages. Its first page stated what the undertaking did not own. Babbage retained the general architecture of the Analytical Engine. The subscribers acquired an interest only in the bounded apparatus, its test, and any receipts earned by its declared use. No success of the apparatus transferred Babbage's later inventions into the venture. No later invention entered the paid configuration merely because Babbage considered it necessary to the truth of the larger design.

"You have put an architectural necessity in the same class as a discretionary enlargement," he said.

"In an account," Halden said, "they both cost money."

They met in a solicitor's room whose table was too small for Babbage's drawings. The inconvenience favoured Halden. Only the configuration register lay open.

Architecture belonged to Babbage. The operational examples, comparison method, and language of any resulting claim required Lovelace's signature. Price, schedule, supplier obligation, and the consequences of an accepted change belonged to Halden. Farrow's works would be paid to manufacture and would be entitled to reject a drawing that supplied no inspectable condition. Guardian retained its source data and the right to accept or refuse the test result. Subscription money would be held separately and released only after four stages: accepted configuration, inspected components, composed assembly, and witnessed test.

The solicitor had left a line beneath Lovelace's name: *deputy, if required*.

"If illness prevents you from attending?" he asked.

"The test waits."

"You could name another signatory."

She held the pen above the blank. "Chosen by whom?"

The solicitor turned a little towards Babbage, waiting for him to speak.

"I have made no objection to your signature," Babbage said.

"No. I know you have not."

She lowered the pen, then lifted it.

"I should like that left out."

The solicitor drew his copy towards him. "The provision?"

"Yes."

She struck it through herself before he could.

Her signature governed the test and its published claim, not the venture's money. Her time and travel still had to be negotiated inside a household whose legal powers were not hers.

The argument lasted longest over change.

Babbage drew a distinction between alterations and corrections. A correction, he said, made the object more nearly what it had always meant to be and should not require the permission demanded by an alteration.

"Who decides which it is?" Halden asked.

"The designer."

"Then we have written sixteen pages to arrive at your memory."

Lovelace set the thin book from Halden's report on the table. "Every proposed change is entered here first. If it enters the present configuration, the affected price, date, interface, and test are accepted again. If not, it remains a successor."

"Successor is the wrong relation," Babbage said. "You make the Engine sound like this instrument's grandson. The later arrangements belong to the Engine; they do not descend from a box of six registers."

He turned several of its numbered leaves. His other hand remained upon the drawing, as if the blank book might remove it. The pages made every future improvement look like a petition to the present.

He left the book open. A change refused here would survive there. Since the government's withdrawal, no one had offered him a refusal that did not also end the undertaking.

The page of funds named three thousand pounds. Babbage supplied half. Outside subscribers supplied one thousand through the trust. Guardian's feasibility and test payments supplied four hundred. Halden supplied one hundred, nearly half his savings.

Lovelace noticed the figure.

"You charge us for refusing risk, Mr Halden, and then you buy some yourself. I should like to know which of you to believe."

"If I didn't, I'd be paid for delay and you for hope."

"And Mr Babbage?"

Halden glanced towards him. Babbage was reading the change clause again.

"Mr Babbage purchases the chance to be contradicted by matter."

Babbage looked up. "It may contradict this arrangement, Mr Halden. I am paying to learn exactly what it contradicts."

"That's why it gets its own signature."

The solicitor turned the agreement to the final page. Halden signed first, because he had drafted the order. Lovelace signed beside the conditions of test and publication. Babbage held the pen while the room grew quiet around him.

"This apparatus," he said, "is not the Analytical Engine."

"That sentence is in the first paragraph," Lovelace said.

"It may nevertheless disclose something true of it."

"That is the question."

He signed. Halden took up the successor register and entered the date. On its first page, Babbage had already pencilled an improvement.

VI. PARTS THAT WOULD NOT BECOME A MACHINE

The two assemblies turned freely until they were joined.

Separate upon the benches, each satisfied its drawing. The gauges entered where they should. The shafts ran true. The report from Farrow's foreman contained no failed dimension. Yet when the shared drive was fitted and the load transferred through both trains, resistance accumulated until three men could not turn the handle without feeling the teeth protest.

Halden arrived at the works before Babbage and found the foreman standing beside the motionless assembly with the guarded expression of a man whose evidence was sound and whose customer would prefer guilt. Farrow stood behind him with the wage book open.

George Farrow had twelve men under wages that week. He owned the lathes through debt and the finished work only until it was accepted. If Babbage called the failure bad manufacture, Farrow would have to defend dimensions whose very success had failed to produce motion. If Halden called it a change, the works could not begin until paper caught up with idle hands. The mechanism's resistance entered Farrow's life first as Friday.

"Uncouple the drive. Give me the first train alone," Halden said.

They uncoupled the drive. The first assembly turned. Then the second. Joined again, they bound before a complete revolution.

"Did either train fail its gauge?" Halden asked.

"No, sir."

"What can you reject?"

"Nothing by itself. Coupled, the drive takes up before one turn. I can show the load. The drawings do not tell me which interface owns it."

Farrow shut the wage book. "Then leave the finding so. I won't have Mr Babbage condemn a sound shaft to make this week's account easier."

Babbage entered while they were repeating the test. He watched half a turn, asked for chalk, and began drawing upon a blackened board. Within minutes he had changed the bearing arrangement, shortened one path, and removed a transfer he had disliked from the beginning.

"I have three accepted interfaces crossed out," Halden said, "a bearing not priced, and a transfer not inspected."

"It removes the cause," Babbage said. "I disliked that transfer from the first drawing, and I have the drawing to prove it. The failure proves those alterations necessary."

Halden tapped the accepted drawing. "It proves this arrangement binds. Farrow can re-machine the interface named here."

Babbage's chalk stopped. "Would you have us reproduce the defect faithfully?"

"I'd have Farrow correct the drive we paid him to make. Your way may be better. It isn't this machine yet."

The workmen had become very interested in tasks that required them to face away from the argument.

Babbage demanded Lovelace's opinion. She was at Ockham and had not seen the assembly. Halden sent her the foreman's measurements, Babbage's drawing, the affected cost pages, and a description of the coupled test. Her answer did not choose the better mechanism.

Could Farrow state a new tolerance for the shared drive and inspect it? If so, she proposed that he try that repair within the accepted arrangement. She could not tell from the report whether Babbage's design would prove the better machine. It should remain in the successor register, with his claim beside it. Neither proposal had yet turned the handle.

"She decides without touching the metal," Babbage said when Halden read the letter aloud.

"She hasn't chosen a bearing," Halden said. "She's told me which repair I can order and still call it a test."

"The distinction becomes very fine when it delays the correct answer."

Halden put both hands on the stationary handle. It did not move.

Farrow's works re-machined the shared interface. The correction cost seventy pounds and eleven weeks. For six of those weeks, Babbage visited less often. His successor drawings grew rapidly.

When the assemblies were joined again, the handle completed one revolution, then ten. The men at the bench looked towards Halden. Halden looked towards Babbage.

The foreman laid two fingers on the bearing cap. At the next turn he removed them and grinned towards nobody in particular. Farrow opened the wage book to enter the date, missed the line, and had to begin again. Halden took the handle himself. Where the load had risen against his shoulder eleven weeks earlier, there was now only the repeating weight of the mechanism. He turned it once more after the test was sufficient.

Babbage put his hand upon the frame and felt the motion pass through it. The vibration climbed through his palm while the bearing gathered heat. The accepted correction had worked. His own improvement might still have been better. The machine offered no judgment on the route not taken.

"Enter both," he said.

The foreman misunderstood and reached for the works ledger.

"Not there," Babbage said. "Lady Lovelace's book. Enter that this turned. And enter what we have deferred."

It was the first time he called the successor register hers.

VII. THE CALL

In October 1847, the third subscription call produced six letters and no money.

Two subscribers requested another month. One offered half. One had failed. One letter came from a solicitor and contained no promise at all. The sixth man explained that under ordinary conditions he remained committed to the undertaking, which was another way of saying that current conditions had released him from commitment.

Halden arranged the letters around the cash account. Farrow's works had earned wages for the completed components. Metal already ordered would fall due the following week. The next contracted stage had not

begun. On paper, the boundary was unusually clear. In the room, it felt like choosing which men would carry the consequence of strangers' fear.

Babbage proposed advancing the next improvement notice.

He unrolled it across two of the unpaid subscription letters. Only the solicitor's seal remained visible.

"Not admitting it," he said. "Circulating it. Men who hesitate before a small apparatus may subscribe to the larger consequence once they see it."

"Then I book their money against machinery we haven't promised and can't price," Halden said.

"Their money would purchase participation in a design whose importance has never been confined to six registers."

Lovelace joined them by letter. Her health had made travel uncertain, and the post laid her hand upon the table in its place.

Her first page kept its usual close hand. On the second the lines stood farther apart, and the three directions—pay, stop, publish nothing—each began a line.

The panic in the money market, she wrote, did not alter what the current subscribers had been shown. If the promise were enlarged when money grew scarce, then scarcity would become a machine for manufacturing claims. Pay the work already earned. Stop before the next stage. Publish neither reassurance nor catastrophe until the account changed.

Halden calculated what delay would do to his own household. His fee for managing the works would stop with them. Margaret could take additional stationer's accounts, but only at night and only if one of the children gave up the room in which she worked. The hundred pounds he had subscribed was already beyond recovery unless the test was reached. Every private consequence pressed towards continuation.

That evening he carried the ledger home.

Their front room served the family after supper and Margaret's stationer's accounts after the children slept. Edward had sometimes described her work as help with his books. She had stopped correcting the phrase. The household knew the difference whenever a customer paid late. That winter, knowledge without a public title was still knowledge that purchased coal.

She moved two school copybooks out of the lamplight and left the coal bill in their place. Edward opened the venture ledger across her account.

"Let me finish this line."

He lifted it. She checked the entry, made her mark, and drew her own book clear before taking his. He kept one hand beneath the ledger until she had room for both pages.

Margaret read the unpaid calls before she read his total. "What happens to the men at Farrow's?"

"They are owed for the stage they finished."

"Can you pay them?"

"If we stop the next stage."

"And if you continue?"

"We may reach the assembly sooner."

She put one fingernail beneath the wages due. "Is this sum in the bank on Friday?"

"No."

He began to explain when the missing calls might arrive. Margaret turned the ledger back to the current balance. After a moment he took the pencil and drew a line beneath it.

Margaret put the accounts aside and took up the small book that had lain beneath her blotter.

"Don't go yet," she said.

"I must find Farrow before he—"

"Not for a minute. Is this the brother?"

She read a passage aloud. It concerned a man whom Edward distinctly remembered drowning.

"That's the same fellow," Edward said.

"No, he drowned."

"I know. You read it to me."

They turned back through several pages. At last she found the other name.

"Here. There were two of them."

"How fortunate for the lady."

"Only one of them has the estate."

She laughed and turned forward again. Edward settled back in his chair. She began a paragraph earlier.

The next morning Halden authorized the earned wages, suspended new manufacture, and extended the schedule without changing the claim. He lost his own fee with the works'. Babbage did not forgive the lost months, but he did not withdraw. The successor register accumulated pages while the accepted apparatus remained still.

By spring, two delayed subscriptions arrived and a replacement subscriber took part of the failed man's call. Manufacture resumed with the same accepted configuration. The delay remained in the account, along with the wages paid before it.

VIII. THE SEALED CASES

The test began with everyone forbidden to improve it.

That prohibition had taken seven years to earn. After the subscription pause, Farrow's men resumed one paid stage at a time. The re-machined interface remained in the present apparatus; Babbage's more general bearing remained in the successor book. By the end of 1850 the six registers, reader, and shared drive composed into one configuration whose parts, order, and exclusions could all be named. Only then did Davies permit the packet to be sealed.

The twelve Guardian cases had been sealed in February. Davies held the source copies and the independently calculated answers. Lovelace held the wording of the question and its exclusions. Halden held the configuration register and the right to call the stop. Farrow's foreman held the inspection record. Babbage held nothing that permitted him to alter the apparatus that morning.

He moved around it restlessly.

The frame occupied less of the room than its reputation did. Six register faces showed through glass. A prepared stack of operation and variable cards stood in its box. There was no printer to dramatize the answer, only a clerk who would transcribe the final values after each run. Visitors expecting a metal intelligence found instead an exact arrangement of limitations. That month half London was going to Hyde Park to see the Exhibition's engines. Nine people had come to watch this one.

Lovelace arrived before Davies. The journey had tired her, though she hid it by studying the seals while the others greeted one another.

She had asked Halden for a chair close enough to the table that she could sit without requesting a pause. It waited by the window before Davies arrived.

"If the first case disagrees," Babbage said, "we must distinguish an error in the supplied number from an error in the operation before we condemn the sequence that follows it. Inspection of the source is not an improvement to the machine."

"After we record the disagreement," she said, and waited for her breath. "Alter the source first, and the answer has chosen its own question."

"A misplaced source figure is not a trial of the Engine."

"Then it is a trial of us."

Davies entered with two clerks and a locked case. He and Halden compared seals. The source values were copied once for the apparatus and once for the room's record. Thomas Neale read each prepared value against Davies's copy before the cards entered the box.

The first attempt consumed the morning, most of it spent by people: each value read aloud and entered by hand, each result transcribed and read back before the next case began. The apparatus itself turned for less than an hour.

Lovelace sat very still and timed each deeper breath to the fall of the card reader. William had opposed the journey and then arranged it carefully when opposition failed. She resented the necessity and relied upon it.

At every transfer, the apparatus made its hard sequence of engagements. It was slow enough that attention wandered and returned. Babbage heard relationships between operations. Farrow's foreman heard bearings. Halden heard paid hours. Lovelace followed the card order with one finger on the written trace, watching the physical machine and the described machine remain the same thing.

The first case agreed. Then the second.

At the sixth, one register failed to clear completely. Halden called the stop before anyone could decide that the remaining digit was harmless. The card was marked and the incomplete run was voided. Davies had already leaned towards the glass. Irritation crossed his face before he told one clerk to note the card and the other to leave the locked case shut. Under observation the register was reset, the inputs were restored, and the full twelve-case sequence began again from its first declared card. The interruption entered the trace; it was not smuggled into either accepted run. Babbage objected to losing the completed cases and then, seeing Davies watch him, objected to nothing further.

The restarted first run consumed the rest of the day. At dusk its twelfth result agreed with Guardian's comparison. The final card struck the receiving box, the shaft ran down, and six register faces held their values in the last light from the window.

No one declared success. The protocol required a second complete run on the following day. The seals were renewed. Halden locked the cards away. Babbage stood beside the silent apparatus after the others had gone.

"The whole declared sequence has agreed once," he said. "The second run tests only repetition; the apparatus has already shown that it can execute the cards."

Lovelace was resting in a chair near the window. "Tomorrow," she said. She did not rise. "That was the condition."

The next morning Davies waited until every seal had been called aloud before he unlocked the independent case; the result sheets remained covered. Neale laid his checking strip beneath the first source line. Halden opened the configuration register at the signed page. Lovelace's chair stood closer to the trace than it had the day before.

Babbage began the second run standing behind the reader. At the third case he moved to the register faces; at the fifth he returned to the cards. He had spent years describing operations whose material sequence no other person had been obliged to wait through. Now he could not make them pass more quickly by explaining them. When he began to tell Davies's younger clerk why the seventh transfer mattered, Lovelace raised one finger without looking up.

"Let him hear the card," she said.

Babbage stopped. The card descended, the transfer engaged, and the register advanced. The clerk looked from the glass to the written operation and found the relation without assistance.

At the ninth case the foreman rested two fingers on the bearing cap. Its warmth had risen evenly. He kept them there through the return of the handle and then nodded once. Halden saw the nod and made no entry for it. There was no line in the account for a load that remained ordinary.

The eleventh case agreed.

For the twelfth, Neale read each prepared value more slowly than before. Davies did not open the expected result. Lovelace's finger reached the final operation and stayed there while the reader took the last card. The shaft completed its turn. One after another the six register faces stopped.

Neale read the value from the apparatus. Only then did Davies uncover the independent sheet.

"Read it again," he said.

Neale did. Davies followed the two columns with the edge of his hand, reached the bottom, and placed the sheet beside the trace. They agreed.

The foreman lifted his fingers from the bearing as though he had been holding the motion there. Halden put down his pencil. Babbage looked first at the registers, then at Lovelace. She was still sitting, one hand closed upon the arm of the chair, but the other remained on the final line. She looked back at him and smiled before either of them spoke.

Davies returned the independent answers to their case and closed it.

Davies signed the comparison statement. Farrow's foreman signed the configuration. Neale signed the preparation check for each run. Halden signed the account of stops and restarts. Lovelace read the public sentence aloud:

The frozen apparatus had executed the declared card sequences upon twelve sealed recurrence cases and had reproduced the independently calculated values on two complete runs under the recorded conditions.

Babbage held out his hand for the sheet. Lovelace kept it upon her lap.

"You have omitted what this demonstrates," Babbage said.

"I read what we signed."

"The separation of operation from value, the direction of mechanism by cards, the general consequence—"

“Add those consequences to your own paper,” she said. “Do not put them above these signatures.”

His hand remained out, but anger failed to arrive. He was overtaken by a joy so close to relief that for several seconds he could not distinguish them. Metal had finally answered beneath his larger design. Men who had called the Engine an evasion would have to speak differently. Halden was shaking Davies’s hand. Farrow’s foreman had begun explaining the restart to one of the clerks. Lovelace held the signed sentence upon her lap.

Farrow’s foreman asked whether he should stop the shaft. Babbage answered too quickly. Lovelace laughed—not at him, but because she had wanted the same thing. The cards were removed, and the empty reader lifted and fell three more times. Halden, who had entered every paid hour of its construction, let it continue. For those three strokes no one watched the register faces.

IX. FOUR DEPOSITS

The first inquiry arrived before the apparatus was covered. A messenger brought a letter from an office that had heard the first day’s rumour and wished to know the price of a calculation. Halden turned the unopened letter over in his hand. It was the first request he had received for something that had not existed when he began to price it.

In July the signed sentence was published without the larger conclusions Babbage had proposed for it. He published those interpretations separately, under his own authority. By December the first inquiry had become four.

The calculation office was constituted in January 1852, in the solicitor’s room where the compact had been signed. Babbage put in six hundred and fifty pounds. Halden put in a hundred and fifty, most of what his household had saved since the first venture, and the hundred pounds left unspent from the construction account made nine hundred. The apparatus came to the office in its frozen configuration. The stopping clauses were copied from the compact, and every customer was to receive two signatures beneath each table: the preparer’s and an independent checker’s.

Lovelace read the articles at Ockham and returned them with a question pencilled beside the fourth, which gave the acceptance of customer work to the manager.

And who may refuse the manager?

Halden wrote beneath it, *The account*, and sent the page back. She signed the article that governed published claims and left the fourth without her name.

By February 1852, Halden’s capacity sheet allowed eighty independent checker-hours in a quarter.

Thomas Neale would not leave an assurance post that carried a pension for an office a month old. His employer spared him ten hours a week, a hundred and thirty in a quarter on paper. But late sources, corrected preparations, customer questions, and ordinary absence consumed the difference. Halden therefore entered eighty independent checker-hours as the quantity he was willing to promise. At an average of forty hours per verified packet, the office could deliver two.

The proposed tables were due between April and June 1854. Customers would close their accounts at the end of 1853, reconcile them through March, and release signed returns no earlier than 1 April. Before then, methods and card sequences could be prepared with specimen or provisional figures. Halden had spread that work over two years. Neale’s final comparison could not begin with figures that had not arrived.

Four offices wished to place deposits.

Neale had no theory of the general Engine. His knowledge was narrower and, at that moment, more decisive. He knew how often a neat source concealed an illegible original, how a correction at the third line required a second reading of the thirtieth, and how fatigue made familiar figures more dangerous than strange ones. None of that appeared in the apparatus’s operation count.

He had cut Wills a checking strip from a rejected card, the twin of his own. When the younger man tried to compare a whole column at once, Neale covered everything below the current line and made him begin again. “A strange figure stops you,” he said, moving the strip down. “Watch the one you expect.”

Wills read the next line and reached for his pencil. Neale kept the strip where it was.

“From the source.”

“I was.”

"You were looking here." He touched the prepared copy.

Wills put the pencil down. At the corresponding place on the specimen sheet the ink had thickened. He knew what the figure ought to be: the other copy gave it plainly. He bent closer to the source, then straightened.

"I can't read that one."

Neale drew the sheet towards the light. For a moment Wills wished he had given the figure after all; Neale was taking longer over it than he had.

"Nor can I," Neale said. "Leave it open."

"Shall I ask for the original?"

"Yes."

Wills marked the place before moving the strip. The rest of the line remained unmarked. He had spent the lesson trying to give Neale no reason to stop him; now he had a question he would have to carry back to the preparation desk. Neale moved his own chair aside so the younger man could get out.

In Halden's new office that February, Babbage held that the estimate described the office they had, while the deposits described the office they might create.

"We are not dividing eighty hours among four fixed objects, as though hours were loaves," he said. "We are using four obligations to purchase more hours. The railways built themselves on exactly that principle, and nobody asks them to apologize."

"From whom?" Lovelace asked.

The office was two rooms adjoining the calculating room. Through the wall came the intermittent motion of the apparatus. On the table lay four letters and the capacity sheet. Lovelace had marked one line twice: independent checking, not preparation, set the boundary.

Halden laid the letters around the capacity sheet. Their corners covered the marked line.

Halden had advertised for another checker. Two men answered; neither could yet challenge a prepared set of cards without Neale's supervision, and training consumed the same scarce hours it was meant to increase.

"The deposits keep Wills paid while Neale trains him," Halden said.

"The customers require checked tables before Wills can sign one," Lovelace said.

"Then we stage the work."

"I know." She reached for the capacity sheet. "I mean the final comparison. When would Neale do it?"

Halden did not answer the question. One letter had called him inventor. He had corrected the word in his reply, but not before reading the salutation twice. Now he looked at the four deposits covering the line.

Babbage stood at the apparatus-room door. "The objection assumes that the present method of checking is fixed for ever, like the tides. A machine exists precisely to free us from the pace of human repetition. I have argued that for twenty years, and I did not expect to argue it in this room."

"It has freed the operation," Lovelace said. "It has not freed us from deciding what operation we asked for. If it had, I should go home and leave you both to it."

"A second apparatus—"

Halden interrupted him. "Isn't in any contract or any account."

Halden looked again at the four deposits.

"Four deposits are not four deliveries," he said. "They pay Wills, retain Neale, and carry the rent while we stagger preparation. We begin training now and accept no fifth."

"Mr Halden has priced every hour by which this office came into being," Babbage said. "He does not mistake a receipt for capacity."

"And Mr Babbage knows which repeated hours the mechanism can remove," Halden said. "We began at opposite ends and arrived at four."

Halden drew the letters inward until their edges made a square. One corner remained over the line Lovelace had marked.

Lovelace asked Neale to join them. He stood awkwardly at the end of the table while Halden described the schedule.

"Four packets at forty hours make a hundred and sixty," she said. "Can you check them independently by the final dates?"

"Not if all four require the hours in the account."

"Could they require less?" Babbage said.

Neale began to answer, stopped, and put the blank strip beneath the total. "Some will. Some will not. I have eighty hours, sir. If I spend them teaching Wills, I cannot enter them again here as his independent check."

No one altered the figure.

Lovelace asked that her objection be entered with the capacity sheet. Babbage would support the four. Neither could sign a customer's acceptance for Halden.

Halden held the office pen. He dated the first acceptance and laid it aside for the second. By the fourth he no longer looked at the capacity sheet. He checked the customer's name, the deposit, and the delivery date. All three were correct. He initialled it and laid down the pen.

During March the four preparations began to overlap. Wills's lessons continued among them.

One afternoon Halden found him beside Neale with the specimen sheets. At the preparation desk a card order waited for a fresh copy. Wills had begun it that morning; another preparer would have to learn the arrangement before finishing it.

"How much longer here?" Halden asked.

Neale kept the strip on the line. "We have not compared his copy yet."

"The cards first, Wills. Then come back."

Wills gathered his pencil and ruler. Neale left the specimen where it was.

By the time the cards were ready, Neale's paid hours for that day were over. Halden put the unfinished lesson beside the next day's work. The card order was complete. It could be entered in the week's account; the lesson could wait for another afternoon. He did not cancel it.

In April Lovelace returned to the office decision. She had not resigned. Both men had feared it, and both would have found it convenient: her absence could have been treated as consent obtained by exhaustion. Instead she took a sheet of the venture's paper and wrote a qualification beneath the February decision. It began in her usual close hand.

The 1851 trial established one frozen operation under comparison fixed beforehand, not the rate at which customer work could be prepared and independently checked. The four obligations depended upon capacity not present when they were accepted.

Below that the lines stood farther apart, and each direction began at the margin.

Accept no fifth.

Pay for Mr Wills's instruction in hours of its own.

Before the first return arrives, name Mr Neale's deputy.

She signed and dated it.

X. THE UNFUNDED SUCCESSOR

By August, Lovelace could work for an hour and lose the afternoon to it.

The arithmetic of her illness was more exact than its cause. A journey to London spent several days. A long letter spent one. Visitors spent whatever strength courtesy concealed from them. At Ockham, papers from the Calculation Office arrived in packets whose urgency was visible before she broke their seals.

William found her with the capacity qualification, Neale's checking sheets, and three letters spread over the bed.

"You promised me one hour," he said.

"I said I would not travel."

"You promised both."

He left the papers where they were. He set the water within her reach and drew the curtains against the light.

"They have made checking depend upon a single man," she said.

"Then it does not depend upon you."

"It depends upon my continuing to insist that he is single."

Her signature stood alone beside the claim clauses in the copy of the compact before her. Beneath it, *deputy, if required* was still legible through the line she had drawn. She set one finger on it.

"Would you have signed it?" she asked.

“What?”

“If they had put your name there.”

“I could not have done your work.”

“No.”

She drew the copy back onto her lap. A moment later she straightened it so he could read the line.

“I would not have let you,” she said.

William sat down beside the bed. She waited for him to say that she had been right. He asked whether the water was still cool.

Wills still learned from Neale between preparations. She had proposed taking him off that work for part of each week: some time under Neale, some with an assurance clerk outside the office. His absence from the preparation desk would have to be paid for as well as the instruction. During training, output would fall further. The additional wages and external review would cost approximately one hundred and twenty pounds in the first year. Halden’s answer praised the plan and asked whether it might begin after the four current packets were delivered. Babbage’s answer proposed that recurrent portions of comparison might later be mechanized.

She read Halden’s answer again, looking for a date she could put beside the cost. Babbage offered a route through machinery that did not yet exist. Neither paper gave her a day when Wills’s hours would be reserved for the instruction.

Lovelace began a page on succession with three headings: competence, paid time, authority to refuse. She entered Wills, Neale, Halden, Babbage, Davies, and herself. No row was complete. She sent the unfinished table to both men and asked them to accept no work beyond Neale’s demonstrated reach until one was.

Babbage came to Ockham near the end of the month. He brought drawings rather than the office account. She recognized the peace offering and the evasion in the same roll of paper.

He opened the roll across her unanswered table. She drew one edge of it back into view.

He showed her a proposed means by which a later apparatus might verify parts of its own operation sequence.

She followed his pencil through two transfers and made him return to the carry. At the next comparison she stopped him again: both checking paths inherited the same earlier value. He stared at the lines, rubbed one away with his thumb, and sent the second path through another register. She moved the delayed sign one operation farther. He saw why before she explained it.

She held the drawing flat where the two paths had inherited the same value. “If a later apparatus compared enough of its own operations, would you call that judgment?”

“Judgment is often the name we give an operation whose conditions have not yet been exposed,” he said. “If the apparatus compares its own proceedings and changes what follows, at what exact line would you deny it the word?”

“I wrote that an Engine might compose music, if the relations of harmony could be given to its operations: that it might weave them, as the loom weaves flowers, into a passage no composer had set down note by note.”

“Would it not have invented the passage?”

“The consequence might be new to us. The relations and purpose would still have entered from us. Our surprise would not be its intention.”

He crossed out *verification* and wrote *comparison of internal operations*.

For several minutes the mechanism occupied the whole table. He began a mark and she completed it; she crossed out his correction and he laughed. Their old speed returned so suddenly that neither acknowledged it. For those minutes William’s hour ceased to exist for both of them. Then she reached for the pencil and had to wait for her breath. When she shifted in the chair, the edge of Guardian’s specimen sheet slipped from beneath the roll.

“It is good,” she said.

He waited for the rest.

“It does not read Guardian’s sheet.”

She drew the successor register from beneath the roll and closed it over his earlier answer.

He did not consent to her table or refuse it. He recommended that the office wait. Halden adopted the recommendation, deferring the paid release and outside instruction until the present contracts produced their next payments.

After he left, Lovelace attached his answer and Halden's to her question. She left the page unclosed, so the unanswered place remained visible.

By November she was in Marylebone, at Great Cumberland Place, and the papers that had once replaced her presence now had difficulty reaching her at all. She died there on the twenty-seventh.

News reached Halden at the office before noon. He closed the preparation room for the day and left the apparatus uncovered. Nobody could say who now held her signature over what the office might claim. Neale stood beside four waiting packets, his blank strip arrested beneath the first line. Babbage came late and went first to the machine. He pressed his palm against the frame and left it there. No motion returned through the metal. It was cold and still.

During 1853 the four packets passed through the office in overlapping stages. On most days this looked like growth. Preparers borrowed desks and reused card forms; Neale moved to whichever promised date was nearest; corrections that once had opened a margin now consumed it. Wills's ruler and pencil moved from packet to packet. The edge of Neale's blank strip darkened with use. No single day announced failure; each exception was carried forward until one could leave the room.

On 1 April 1854 Guardian's signed return arrived, and the Lothbury office's with it. Wills laid Guardian's beside the specimen sheets. Much of the card order could be kept. Neale put the old checking sheets aside; their ticks belonged to different figures. The Lothbury copy was ready first, and he drew its first line beneath his strip. Lothbury's and Guardian's were the two small packets, a week of Neale's hours apiece; the two large ones would need the rest of the quarter and more.

On Friday the seventh both dates fell due, the first the office had ever had to keep.

The Guardian cards were ready a little after nine. Wills brought the cards and his copy to Halden's desk. There was one mark in the checking column, his own.

"Mr Neale?" Halden asked.

"The Lothbury comparison, sir. He hopes to finish it by four."

Through the glass Neale sat with his strip beneath a line of another office's figures. He had perhaps six of his week's hours left. Halden could walk twenty feet and put the Guardian copy on top of the Lothbury sheets, and one of the two offices would receive its table late.

The wage envelopes lay filled in their box on the corner of the desk. Guardian's deposit had paid for some of them; if the date passed, the deposit became returnable. On the shelf above, beside the test packet, lay Lady Lovelace's qualification.

He opened the delivery ledger at the Guardian line. No line in it had yet been marked delivered. He wrote in the margin reserved for an exception.

Second comparison to follow. Delivery permitted to preserve date.

E. H.

"Release them," he said. "Mr Neale will compare the copy on Monday."

Wills looked at the single mark and then at the margin. He squared the cards against his palm and carried them to the reader.

The cards began to fall at ten. By four the table had gone to Guardian in Lombard Street, and Lothbury's had followed it with both signatures and a quarter of an hour to spare.

On Monday, with the papers full of the war, the third office's signed return arrived with a query in its covering letter, and Neale's hours went to the query. Halden had meant to tell him about Guardian's copy once the query was answered. On Tuesday the fourth arrived. Neale worked, as he had all year, to the nearest promised date, and a delivered table had none. The Guardian line in the ledger carried its single mark forward from week to week, as the year's exceptions had, until the packet came back tied with the office's own green tape.

XI. THE SELLER'S CONDITIONS

Halden's reply to Guardian contained the source sheet, the preparation copy, the cards, both run traces, the page of the delivery ledger, and the sentence with which he had begun: there was no second signature.

Davies accepted the trace and rejected the table. "I do not doubt your mechanism," he wrote. "Guardian did not purchase it."

Clause eleven required correction at the seller's expense. The following clause permitted cancellation if the error had passed a promised independent check or if no such check had occurred. Guardian invoked both clauses. Its deposit of one hundred and eighty pounds was to be returned. The corrected table would be received as evidence of performance under the clause, not as completion of a continuing contract.

Neale had come in at two on the day the packet returned. He read the margin of the delivery ledger before he took off his coat, and Wills told him what had been said at the desk that Friday.

"You promised my Monday, sir."

"I did."

Neale hung the coat on its peg and asked for the source sheet.

The office recomputed the table. Wills copied the source under Neale's voice, then Neale copied it again under Wills's. The apparatus produced the corrected values. The work took four days. By the third evening Neale's voice had roughened, and Wills read while Neale followed with the grey-edged strip.

When Neale stopped him, Wills began the whole line again, faster.

"From where the strip is."

Wills found the place. He read the figure once. Neale moved the strip, and they went on.

When they exchanged sheets, Wills asked for a repetition. The request came out too softly; Neale had already begun the next figure. Wills put his pencil across the copy.

"The one before, Mr Neale."

Neale returned to it. Wills entered the figure only after he had heard it again. Neither man looked towards the office door. There was still the rest of the column to read.

The correction had required no alteration to the mechanism. Babbage wanted the response to say so.

He slid the two exact run traces away from the delivery ledger. Halden returned them to the same packet before answering.

"Put this before the refund," Babbage said. "The difference enters before the first card."

"The trace makes it clear," Halden said.

"A trace in a cancelled office will not govern what people say. They will say the Engine made a wrong table. I began the first engine because tables computed and copied by hand were wrong, Mr Halden. I gave it a printer so that no hand need copy a figure between the machine and the page, and you struck the printer from your first report."

"The figure went in by hand. A printer would have printed it."

"I know it," Babbage said. "I have twice been asked, by members of both Houses, whether the machine gives right answers when it is given wrong figures. They will not need to ask again. They will think this table has answered them."

"What did Guardian purchase?" Halden said. "It paid us to stop that table leaving. The clerk has his line in the trace. Mine is in the ledger. I won't answer cancellation with a paragraph about the Engine."

Halden opened the office account.

There had been nine hundred pounds in contributed capital and seven hundred and twenty in customer deposits. Wages, rent, heat, cards, maintenance, and delivery had consumed most of it. Guardian's refund, the correction, reprinting, closure, and storage consumed the rest. Even if another customer agreed to wait, the office could not support the staff required to complete and independently check the remaining work without new capital.

"Then raise new capital on the investigation," Babbage said.

"On what statement?"

"That the calculating mechanism executed every operation supplied to it; preparation and checking failed outside it."

"And the office?"

"Those functions can be constituted anew under stricter conditions."

Halden imagined the circular. A correct machine, a corrected human process, subscriptions sought for renewed operation under improved safeguards. Every sentence could be defended. Together they would invite a new subscriber to finance obligations the present account could not meet.

Margaret had sorted the figures the night before into sums already owed, money still in hand, and money a new circular only hoped to attract. When Halden began to call the third column a rescue, she had put one finger beneath the first.

"Before you send any circular," she had said, "show me what leaves this house before the first new subscription comes in."

She had found his capacity sheet among the office papers, the line marked twice. "You showed me the articles in January, when you asked for the savings," she had said. "Not this."

He had not.

New money might preserve Halden's standing and keep Wills and Neale employed. He looked again at the sums already owed.

He wrote the closure resolution.

The remaining customer obligations would be completed only where existing funds and independent checking sufficed; otherwise they would be cancelled before another stage became due. Guardian would receive its refund and corrected table. Wills and Neale would be paid through the closure month. The apparatus would be dismantled into stored assemblies. The complete ledger would name the February 1852 four-job authorization and the April 1854 delivery exception. Under *cause of exception*, Halden first wrote *pressure of business*. He read the words, struck them through, and copied both of his authorizations in full.

Wills asked to see him after the staff were told.

"Will the record carry my name?" the young man asked.

"On the preparation copy."

"And on my reference?"

"The reference will state both the error and what you did afterward."

Wills looked towards the preparation room. "A new master may stop at the error."

Neale had waited at the door with his wooden case. "Put my name beneath the correction, sir. He read the second copy. I followed."

Halden added the sentence. Neale initialled it, then Halden turned the reference around and placed it beside Wills's ruler. "Read it before I sign."

Wills put the ruler beneath the first line. He read down to Neale's initials, moved it back to the top, and began again.

Neale lowered his wooden case to the floor.

"Is there something wrong?" Halden asked.

"Could you put the order in, sir?" Wills asked.

"The order?"

"To send the table before the second comparison."

Halden brought the pen towards the first paragraph.

"Not the copying," Wills said. "I did that."

"I wasn't going to take it out."

Wills looked at the ruler. "Only the other part as well."

There was room beneath Neale's initials if Halden wrote small. He drew a fresh sheet from the drawer instead.

He copied the account of the transposition, then wrote: *I authorized delivery before the independent comparison was complete*. Below it he put the account of the correction.

Neale bent over the new sheet. His case stood between his feet while he initialled his sentence again.

Wills put the ruler beneath the first line. When he came to the new sentence he read it twice. Then he moved the ruler down.

The remaining obligations took until the spring to close. On the last day of March 1855, the shaft beneath the calculating room stopped. Farrow's men removed the card reader first, then the registers. Neale carried his checking books away in a wooden case. Wills left with a reference that described both the error and his conduct during correction.

Halden locked the account and gave one key to the subscriber trust's solicitor and one to Babbage.

"You have used the stopping clauses as though dissolution were a completed stage," Babbage said.

"The wages are paid. The customers have their money or their work. One key goes to the trust and one to you."

"That preserves the account by surrendering the work."

Halden waited. Babbage closed his hand around the key.

"It does," Halden said.

XII. THE COMPLETE RECORD

In 1871, Henry Babbage found the returned table filed with his father's successes.

It had remained there since the office closed, sixteen years. At first Charles pencilled *mechanism trace exact* upon the wrapper. Later he asked Halden for the reconciled account and tied it beneath the same green tape. The tape had been opened often enough to soften at the knot.

Charles had asked him to inventory what would pass into his custody. The work occupied several afternoons of that last autumn. Drawings had multiplied beyond cabinets. Mechanisms stood beneath covers. Notes on improvements continued into the present year, written in the same hand that had marked changes before Henry was born.

The packet from the old undertaking was different. It had been assembled as if for a dispute whose parties might all be dead. The frozen configuration came first, then the successor register, inspection corrections, sealed cases, run traces, the second smudged at its last operation, and the signed statement of 1851. After them lay Lady Lovelace's capacity qualification, her unanswered succession question, the source sheet, the transposed copy, the delivery exception, Davies's cancellation, the corrected table, and Halden's closure account.

Her qualification began in the close hand Henry knew from her letters to his father. Its final directions stood farther apart, each beginning at the margin.

Henry lifted the returned table. "This is filed with the Engine papers."

His father was seated near the fire, wrapped in a coat although the day was not cold. Age had reduced the distance he could cross without effort but not the distance his thought attempted in a sentence.

"It was not an error of the Engine," Charles said.

Henry set it beside the office accounts.

"Not there," Charles said.

Henry left the sheet between the two piles. "Then the inventory needs a heading that takes both."

His father looked from one pile to the other.

"That was Halden's question," Charles said at last. "What had the customer purchased?"

Henry kept his pencil above the blank line.

"I distinguished the general machine from a temporary arrangement of persons," Charles looked towards the covered mechanisms. "The distinction was true."

Henry waited.

"But they bought the table," Charles said.

Henry had grown up among unfinished objects defended by future completion. Even now the covered mechanisms drew his eye away from the account. There was work he could imagine doing with them, and he had to bring himself back to the blank inventory line. The returned table still lay between the two piles. Whatever happened to the drawings, it would need a place.

Military service had taken Henry far from these rooms. Now he had returned, with questions of his own about the machinery, and his father tired before they could finish an argument.

Charles noticed him looking towards the drawings.

"Not that sheet. The alteration is on the next."

Henry lifted it. Underneath lay a drawing of the mill, its carriage crowded with pencilled corrections of several dates.

"Here?"

"One farther."

Henry carried the drawing to him. They followed a line together until it passed beneath another sheet. His father was still tracing its continuation when Henry moved the paper aside.

"Yes," Charles said. "You see."

Henry returned the drawing to its place and took up the pencil beside the inventory.

"What name shall I put upon the inventory?" Henry asked.

Charles answered too quickly. "Analytical Engine—experimental working papers."

Henry wrote the phrase, then left space beneath it.

His father noticed. "You disapprove?"

"It does not explain why the accounts are present."

Charles asked for the inventory. His hand was unsteady, and Henry expected him to strike out the first line. Instead he added another.

Bounded apparatus and calculation office: configuration, proof, qualification, error, correction, and closure, complete.

“Complete,” Henry read.

“Of that object,” Charles said. “Not of the Engine.”

“I understand.”

“Do you?”

The old impatience sharpened the words, but it could not sustain itself. Charles rested the pen.

“No matter. Keep the sentence.”

The packet passed into Henry’s custody with the drawings and workshop material after his father’s death. He copied both headings into the inventory and placed the packet beside the drawings.

In 1900, deciding what should pass after him, Henry opened the packet again.

The green tape had faded. The wrong table and the corrected one lay side by side. Between them was Lovelace’s qualification, its final lines still widening towards the end.

Henry returned her sheet between the two tables and tied the faded tape around all three.